Sales Data Analysis Project

Exploratory Data Analysis

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Tools Used: Python, Pandas, Matplotlib

Objective: Exploratory Data Analysis & Business Insights

Executive Summary

This project analyses transactional sales data to uncover meaningful insights related to revenue trends, customer behaviour, regional performance, and payment preferences.

The analysis involved data cleaning, feature engineering, and exploratory data analysis (EDA) using Python. Key findings include seasonal revenue fluctuations, strong regional performance in the West, digital payment dominance, and opportunities to increase Average Order Value (AOV) through strategic initiatives.

The objective of this project was not only to analyse data but also to translate insights into actionable business recommendation

Dataset Description

Overview

The dataset consists of transactional sales records where each row represents a single customer order. The dataset captures information about customers, order details, region, payment methods, and pricing.

Dataset Structure

The dataset includes:

- Order_ID
- Customer_ID
- Customer_Name
- Order_Date
- Region
- Payment_Method
- Quantity
- Unit_Price

Dataset Size

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10200 entries, 0 to 10199
Data columns (total 13 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   Order_ID        10200 non-null  object
 1   Order_Date      10200 non-null  object
 2   Customer_ID     10200 non-null  object
 3   Customer_Name   9999 non-null   object
 4   Category        10200 non-null  object
 5   Sub_Category    10200 non-null  object
 6   Quantity        10200 non-null  object
 7   Unit_Price      10200 non-null  object
 8   Discount        10200 non-null  float64
 9   Sales           10200 non-null  float64
 10  Profit          10200 non-null  float64
 11  Region          9998 non-null   object
 12  Payment_Method  9995 non-null   object
dtypes: float64(3), object(10)
memory usage: 1.0+ MB
```

Data Cleaning & Preprocessing

Initial Data Observations

- Dataset contains 10,200 rows and 13 columns.
- 3 numerical columns: Discount, Sales, Profit.
- 10 object columns.
- Quantity and Unit_Price are incorrectly stored as objects.
- Order_Date is stored as an object (should be a datetime).
- Missing values present in:
    a. Customer_Name (201)
    b. Region (202)
    c. Payment_Method (205)
- Duplicate rows likely exist (dataset was intentionally created with duplicates).

Checking Missing Values

```
df.isnull().sum()
```

```
Order_ID            0
Order_Date          0
Customer_ID         0
Customer_Name     201
Category            0
Sub_Category        0
Quantity            0
Unit_Price          0
Discount            0
Sales               0
Profit              0
Region            202
Payment_Method    205
dtype: int64
```

Handling Missing Values

```
# Handling Missing Categorical Values

df.loc[:, 'Customer_Name'] = df['Customer_Name'].fillna("Unknown")
```

```
# Filling the missing values with the most frequent Value(mode)

df.loc[:,'Region'] = df['Region'].fillna(df['Region'].mode()[0])
```

```
# Filling the missing values with the most frequent Value(mode)

df.loc[:, 'Payment_Method'] = df['Payment_Method'].fillna(df['Payment_Method'].mode()[0])
```

Checking and Removing the Null Values

```
# Checking for null values

df[['Quantity','Unit_Price']].isnull().sum()
```

```
Quantity       104
Unit_Price     100
dtype: int64
```

```
# Removing the rows with null values

df = df.dropna(subset=['Quantity','Unit_Price'])
```

Fixing Inconsistencies

```
# Fixing Text Inconsistencies

df['Region'].unique()
```

```
array(['South', 'East', 'West', 'north', 'Central'], dtype=object)
```

```
df.loc[:,'Region'] = df['Region'].replace('north','North')
```

Removing Duplicate values

```
# Removing Duplicate Rows

df.duplicated().sum()
```

```
np.int64(179)
```

```
df = df.drop_duplicates()
```

Feature Engineering

Creating a Revenue column

```
# Creating a Revenue Column

df['Revenue'] = df['Quantity'] * df['Unit_Price']
```

```
df[['Quantity','Unit_Price','Revenue']].head()
```

	Quantity	Unit_Price	Revenue
0	2.0	880.21	1760.42
1	2.0	592.03	1184.06
2	4.0	884.42	3537.68
3	9.0	236.08	2124.72
4	8.0	208.63	1669.04

Extracting Year & Month

```
# Extracting Year & Month

df['Year'] = df['Order_Date'].dt.year
df['Month'] = df['Order_Date'].dt.month
```

```
df['Month_Name'] = df['Order_Date'].dt.month_name()
```

Key Performance Indicators (KPIs)

```
# Business Metrics

# Total Revenue

df['Revenue'].sum()
```

```
np.float64(25857601.65)
```

```
# Total Order

df['Order_ID'].nunique()
```

```
9709
```

```
# Total Customers

df['Customer_ID'].nunique()
```

```
1000
```

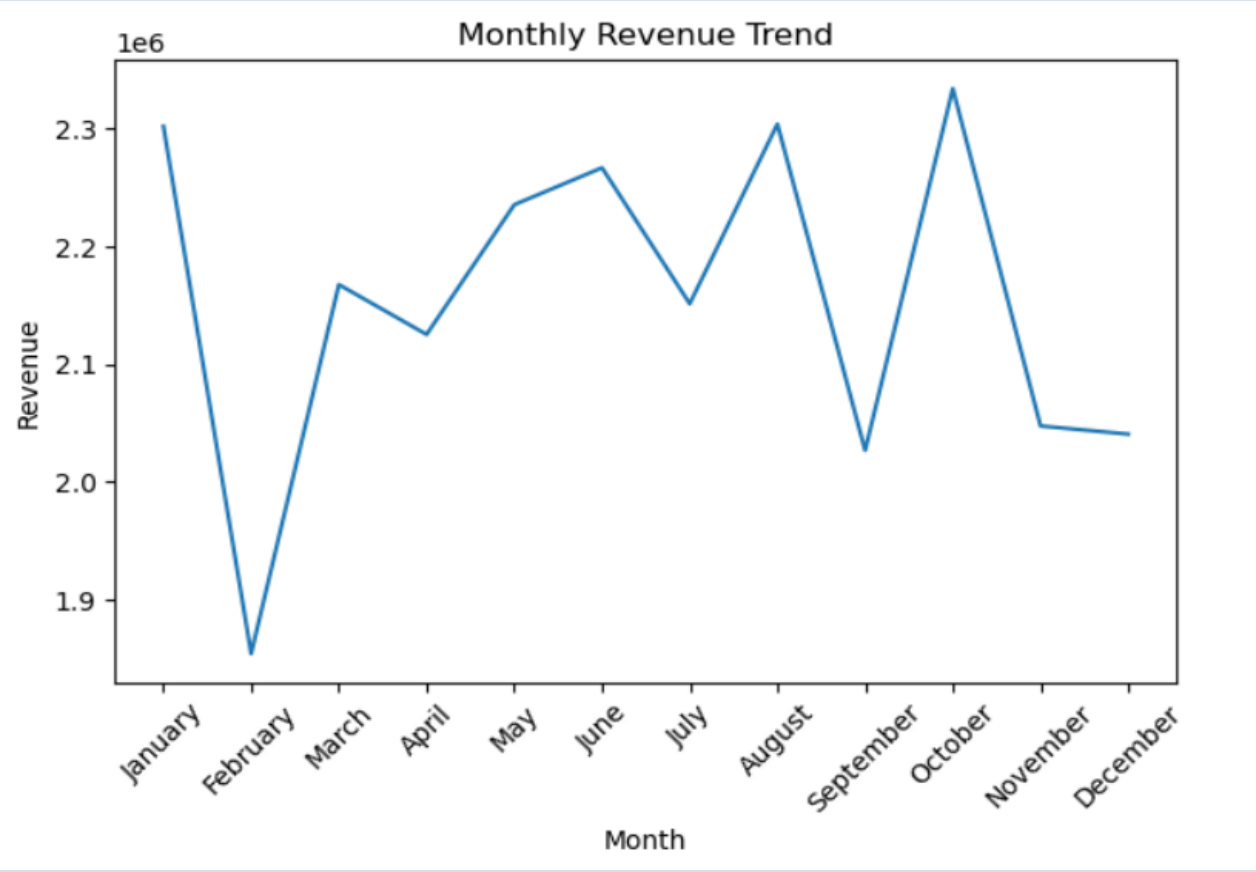
```
# Average Order value (AOV)

df.groupby('Order_ID')['Revenue'].sum().mean().round(2)
```

```
np.float64(2663.26)
```

- **Total Revenue: ₹25.85M**
- **Average Order Value: ₹2663.26**
- **Total Orders: 9709**
- **Total Customers: 1000**

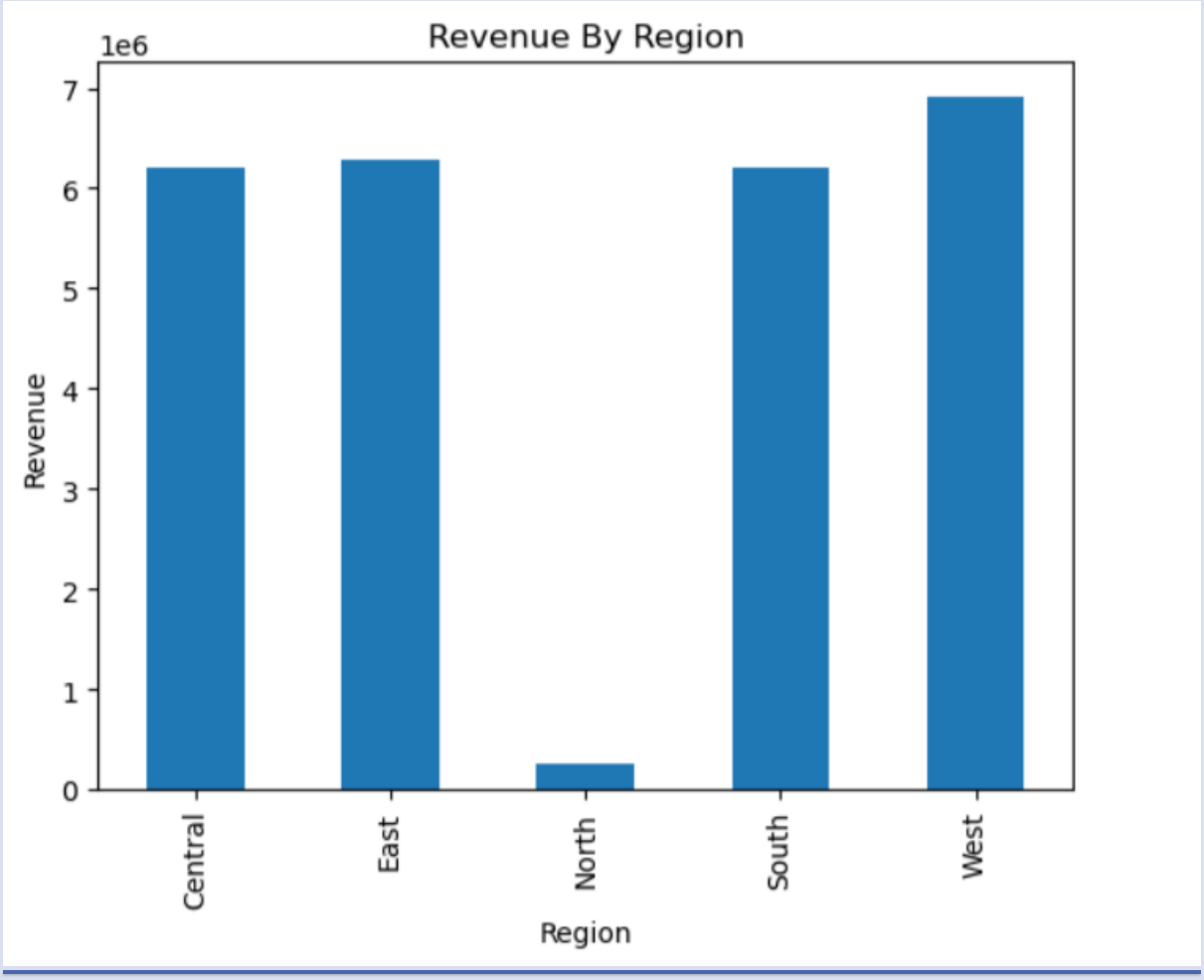
Exploratory Data Analysis (EDA)

Monthly Revenue Trend



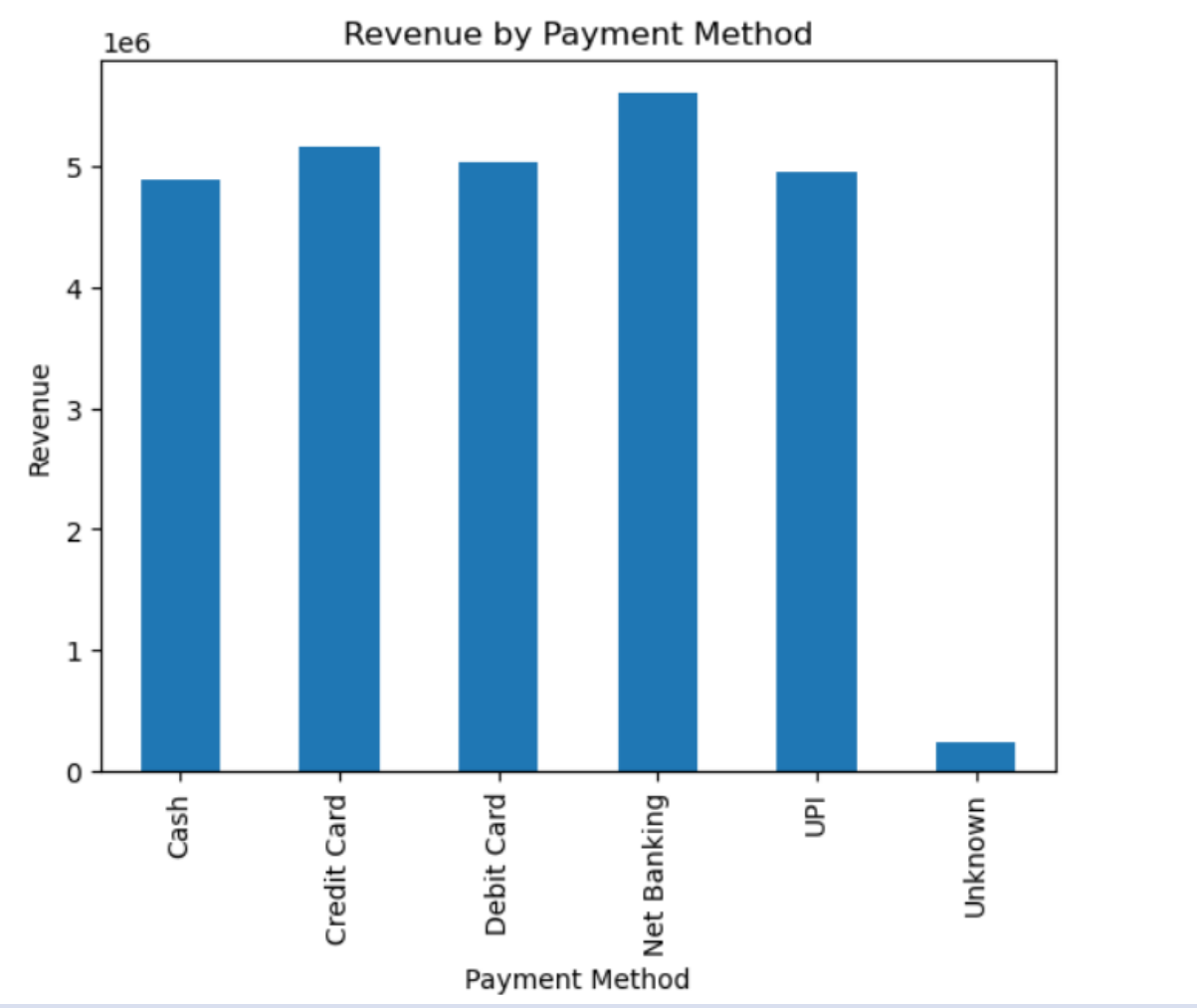

```
# Monthly Revenue Trend Chart

monthly_revenue = (
    df.groupby(['Month', 'Month_Name'])['Revenue']
      .sum()
      .reset_index()
      .sort_values('Month')   # Ensures Jan → Dec order
)

plt.figure()
plt.plot(monthly_revenue['Month_Name'], monthly_revenue['Revenue'])

plt.title("Monthly Revenue Trend")
plt.xlabel("Month")
plt.ylabel("Revenue")
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

Revenue by Region



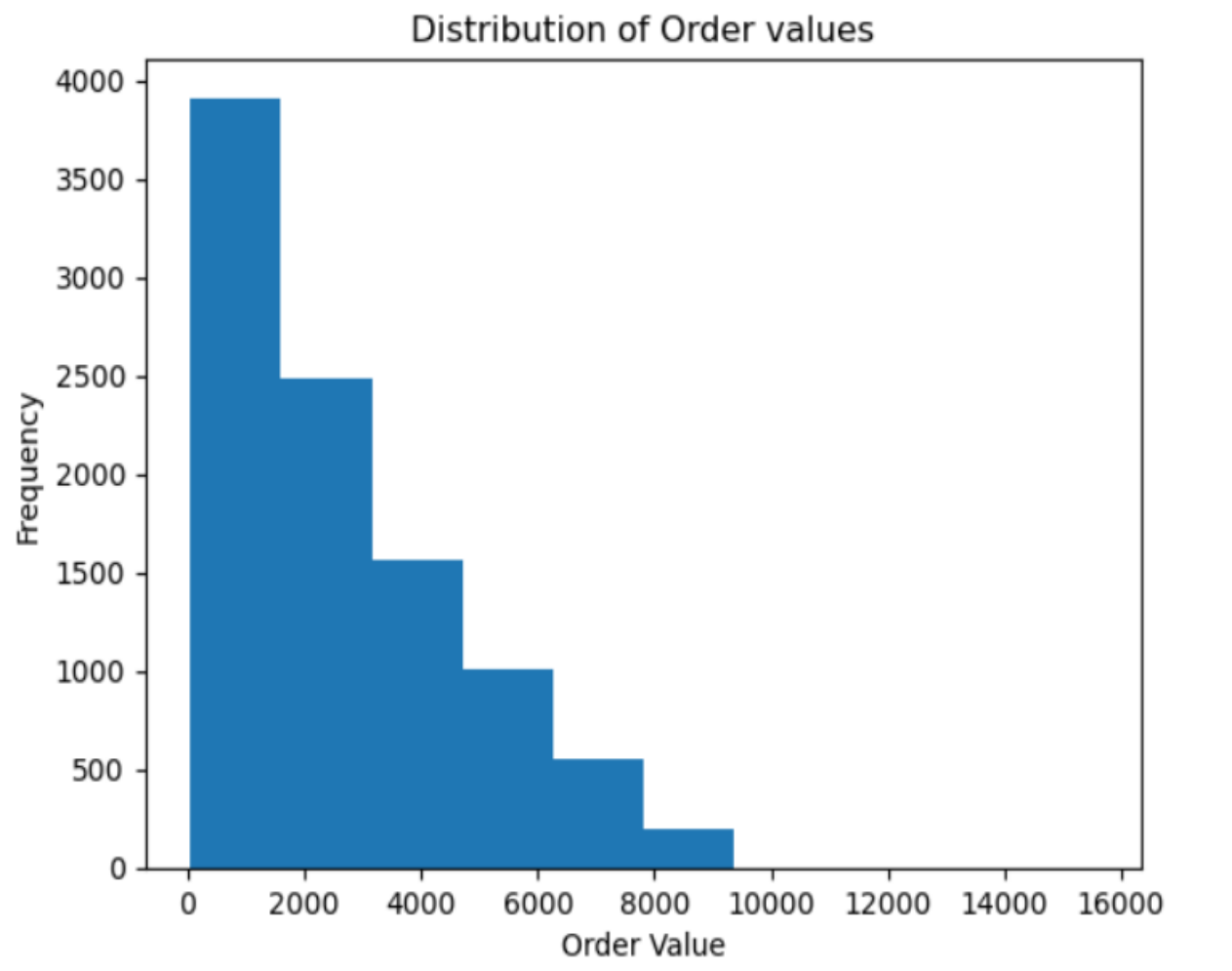

```
# Revenue By Region Chart

region_revenue = df.groupby('Region')['Revenue'].sum()

plt.figure()
region_revenue.plot(kind='bar')
plt.title("Revenue By Region")
plt.xlabel("Region")
plt.ylabel("Revenue")
plt.show()
```

Revenue by Payment Method



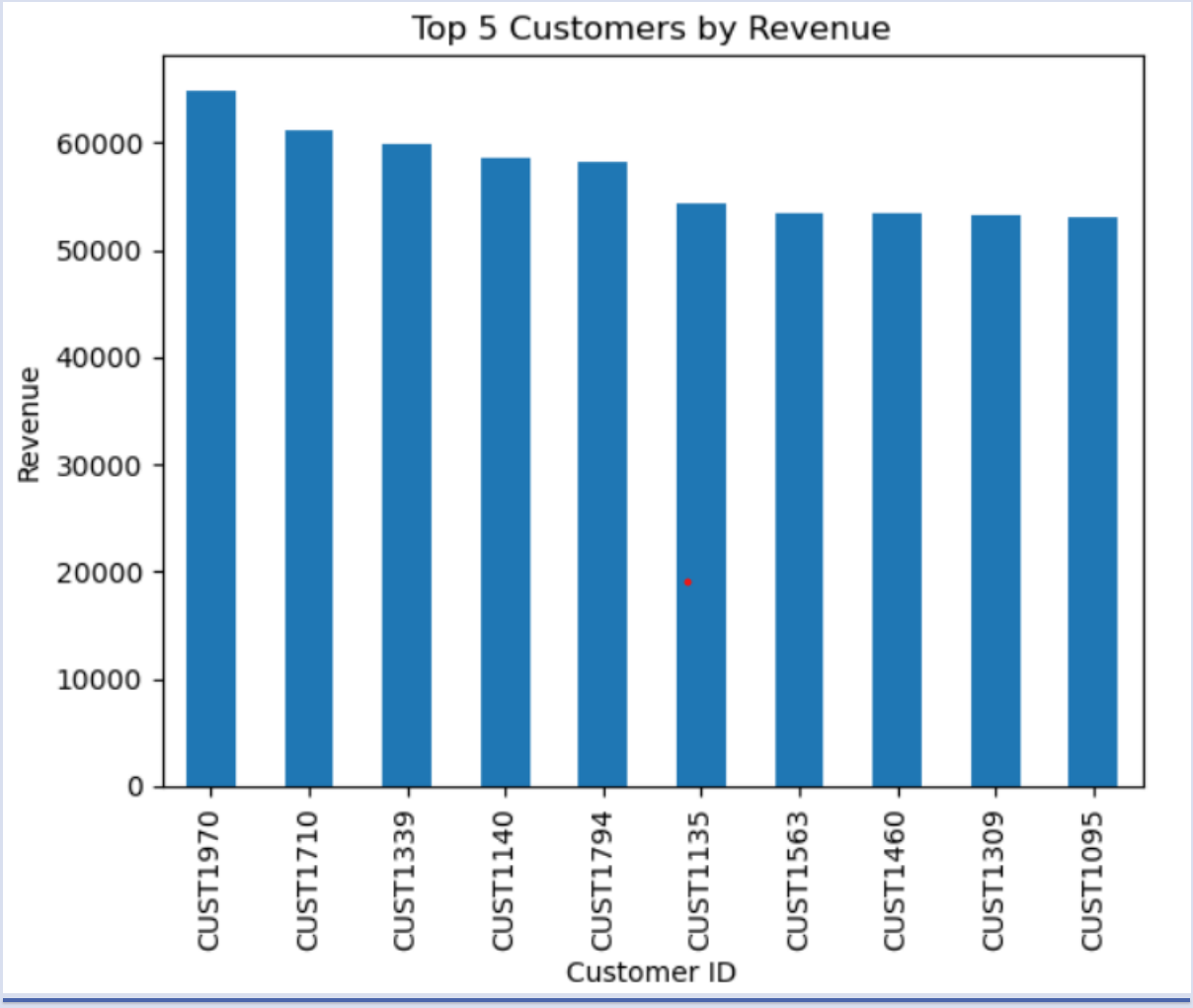

```
# Revenue by Payment Method chart

payment_revenue = df.groupby('Payment_Method')['Revenue'].sum()

plt.figure()
payment_revenue.plot(kind='bar')
plt.title("Revenue by Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Revenue")
plt.show()
```

Order Value Distribution




```
# Order Value Distribution Chart

order_value = df.groupby('Order_ID')['Revenue'].sum()

plt.figure()
plt.hist(order_value)
plt.title('Distribution of Order values')
plt.xlabel('Order Value')
plt.ylabel('Frequency')
plt.show()
```

Top 10 Customers by Revenue




```
# Top 10 Customers by Revenue Chart

top_customers = (df.groupby('Customer_ID')['Revenue'].sum().sort_values(ascending=False).head(10))

plt.figure()
top_customers.plot(kind='bar')
plt.title("Top 10 Customers by Revenue")
plt.xlabel("Customer ID")
plt.ylabel("Revenue")
plt.show()
```

Business Insights Summary

- Revenue demonstrates an overall upward trend during March, May, June, August, and October, indicating periods of strong customer demand and positive market performance.
- However, revenue declines are observed in February, April, July, September, and November, suggesting potential seasonal slowdowns or operational inefficiencies. Further investigation into marketing campaigns, pricing strategies, or external factors during these months is recommended.
- The West Region contributes the highest share of total revenue, indicating strong market penetration and brand presence in this area.
- Expanding targeted marketing and promotional strategies in the North Region could help diversify revenue streams and reduce regional dependency risk.
- The dominance of digital payment methods highlights a strong customer preference for convenience and cashless transactions.
- Maintaining a secure, seamless, and optimized digital payment infrastructure is essential to sustain customer satisfaction.
- Revenue is not overly concentrated among a small group of customers, indicating a healthy and diversified customer base. This reduces revenue risk associated with customer dependency.
- The majority of orders fall within the lower-value range, presenting opportunities for upselling, bundling, and cross-selling strategies to increase Average Order Value (AOV) and overall profitability.

Business Recommendations

- The company can introduce discounts, festive offers, or special campaigns during the months of February, April, July, September, and November to improve sales. It would also be helpful to check if there were fewer marketing activities or external factors affecting sales during these periods.
- Since customer demand is higher in these months, the company can increase product stock, run premium campaigns, and promote best-selling products to maximize profits.
- While maintaining strong performance in the West, the company should focus more on improving sales in other regions like the North. Targeted ads and regional offers could help balance revenue across regions.
- The company should ensure smooth and secure digital payment options. Offering small rewards or cashback for digital payments can further encourage customers.
- The company can increase revenue by:
    1. Offering product bundles
    2. Suggesting related products at checkout
    3. Providing free shipping above a certain amount

CONCLUSION

Through this project, I analysed monthly revenue trends, regional performance, payment methods, and customer purchase behaviour. The data shows that the business is performing well overall, especially in certain months and regions. However, there are opportunities to improve revenue stability and increase order value.

By using targeted marketing strategies, expanding regional focus, and improving customer experience, the company can achieve more consistent and sustainable growth.

This project helped me understand how data analysis can provide meaningful insights that support business decision-making.